Relative Performance Feedback and Redistribution Preferences: Evidence from an Online Experiment*

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Abstract

This paper examines how relative performance feedback affects redistribution preferences and beliefs about effort versus luck in generating inequality. In an online experiment, participants complete a code-recognition task and are randomly assigned to one of three conditions: no feedback, told they performed *above* a randomly matched peer, or told they performed *below*. They then act as impartial spectators and redistribute a fixed sum between two workers whose earnings reflect their performance. Participants told they outperformed their peer score 1.29 points higher on an effort-attribution scale (0–10) than those told they underperformed ($p < 0.05$) and redistribute \$0.57–0.65 less to the lower-earning worker ($p < 0.05$). The no-feedback group sits between the two treatment arms, serving as an uninformed baseline anchor. For redistribution it does not differ significantly from either treated group; for effort beliefs, however, the below group scores significantly lower than the no-feedback anchor ($p < 0.10$), consistent with negative feedback having a stronger belief-shifting effect than positive feedback. A mediation analysis shows that roughly 45 percent of the redistribution effect is explained by the shift in effort–luck beliefs, consistent with self-serving attribution as the channel linking feedback to redistribution. As a novel design element, participants report their expected relative performance before receiving the feedback signal; we find that 85 percent predicted to outperform, reflecting a strong optimism bias that constrains identification of the surprise channel in this sample.

*This experiment builds on the design of Almås et al. (2020) and is inspired by the research question of Cassar and Klein (2019).

Keywords: redistribution, relative performance feedback, self-serving attribution, fairness, online experiment
JEL codes: C91, D31, D63, D91

1 Introduction

Free-market economies inherently generate unequal outcomes, and while modern democratic societies rely on redistribution to mitigate these disparities, there is persistent disagreement over which inequalities are considered legitimate (Alesina and Giuliano, 2011; Piketty, 2014). A central dimension of this debate concerns the *source* of inequality: outcomes attributed to effort are typically viewed as more deserving than those arising from luck. This intuition is formalised in the accountability framework, which models redistributive preferences as judgments about deservingness and predicts lower redistribution when inequality reflects effort rather than chance (Konow, 2000; Cappelen et al., 2007). The framework finds broad empirical support: beliefs about the determinants of income are consistently among the strongest predictors of redistribution preferences (Piketty, 1995; Alesina and Angeletos, 2005; Bénabou and Tirole, 2006).

Yet a key difficulty in this debate is that, in practice, the origins of inequality are rarely clear-cut. Individual outcomes typically reflect a combination of effort, ability, family background, and luck, and it is often hard to separate these contributions, even for the person directly involved (Piketty, 2014). This ambiguity creates room for motivated reasoning[1] (Kunda, 1990): when outcomes cannot be unambiguously attributed, beliefs about their causes are shaped not only by available evidence but also by personal experience and the desire to see oneself in a positive light — systematically tilting fairness judgments, and through them, support for redistribution.

Deffains et al. (2016) argue that the experience of success and failure shapes redistribution preferences through the self-serving bias: individuals who succeed at a task attribute their outcome to effort and favour less redistribution, while those who fail attribute their outcome to circumstance and shift toward more. Since wealthier individuals have typically experienced more success, this bias systematically pushes them to view inequality as earned rather than lucky, creating a divide between rich and poor not just in material interests but also in fairness views. As Deffains et al. (2016) put it, wealthy individuals "do not only oppose redistribution because they expect to be net payers, but also because the self-serving bias systematically shifts their fairness principles." Identifying this channel outside a controlled setting is difficult, however: in observational data one cannot tell whether successful individuals attribute outcomes to effort because of cognitive bias or because they genuinely worked harder. Isolating the self-serving bias therefore requires exogenous variation in comparative signals, holding actual performance constant while shifting only what individuals

[1] Motivated reasoning refers to the tendency to process information in ways that support preferred conclusions or protect self-image, rather than neutrally evaluating evidence.

learn about their relative standing. This paper asks whether such a signal — being told one outperformed or underperformed a peer — is enough to shift beliefs about the role of effort in generating inequality and, through that channel, redistribution preferences.

This paper provides experimental evidence on this mechanism. Participants complete a performance task and are randomly assigned to one of three conditions: (i) *no feedback* about relative standing; (ii) told they performed *above* a randomly matched peer (*above* group); or (iii) told they performed *below* a matched peer (*below* group). Subsequently, all participants act as impartial spectators and decide how to distribute a fixed monetary sum between two workers whose inequality arose from differential performance.

The key findings are four. First, participants told they outperformed a peer redistribute less in the merit scenario: \$1.92 on average, versus \$2.49 in the below group and \$2.08 in the no-feedback group. The above–below gap of \$0.57–0.65 is significant ($p < 0.05$) across both specifications. Second, the same group scores 1.29 points higher on an effort-attribution scale (0–10) than the below group ($p < 0.05$), consistent with self-serving belief updating; the joint test across all three groups also rejects equality of means ($p = 0.041$, ANOVA). A mediation analysis shows that roughly 45 percent of the redistribution gap is accounted for by the shift in effort–luck beliefs, providing direct evidence that self-serving attribution is the operative channel linking the comparative signal to redistribution. Third, the no-feedback group anchors in the middle of both outcomes. We include this arm specifically to ask whether the *information content* of the comparative signal — not task completion alone — is what drives belief updating. Its intermediate position is consistent with that interpretation: both comparative signals push beliefs and redistribution away from an uninformed baseline in opposite directions. The significance pattern differs across outcomes. For redistribution, neither treated group individually differs from the no-feedback anchor ($p = 0.26$ for the below–no-feedback comparison), and the above–below gap is the only significant contrast. For effort beliefs, the below group scores significantly lower than the no-feedback group ($p < 0.10$), while the above–no-feedback gap does not reach conventional significance, suggesting that negative feedback shifts merit attributions more forcefully than positive feedback; the most precisely estimated contrast remains the above–below divergence. Fourth, by classifying participants according to whether their feedback confirmed or contradicted their pre-feedback prediction, we identify a *confirmation* channel. Among participants who expected to outperform their peer, those whose expectation was confirmed (*positively reassured*: told above, predicted above; $n = 42$) score 1.16 points higher on effort attribution and redistribute \$0.63 less than those whose expectation was contradicted (*negatively surprised*: told below, predicted above; $n = 37$), with both contrasts significant at the 5 percent level. This is the sharpest contrast in the data: confirmation of a prior belief in one's own superiority triggers the strongest merit-oriented response, consistent with motivated reasoning amplifying the self-serving attribution effect when feedback reinforces rather than challenges an existing self-view.

These results are consistent with the self-serving attribution account developed by Deffains et al. (2016): individuals who perform well relative to others update their beliefs about the role of effort in generating inequality and become less redistributive. Our design extends their framework by randomly assigning the comparative signal, providing clean causal identification and isolating the relative — rather than absolute — performance channel. The paper most directly relates to Almås et al. (2020), from whose spectator design we adapt the merit redistribution task, and to Cassar and Klein (2019), whose focus on how performance comparisons shape distributive preferences motivates our research question.

A second contribution is a novel design element: before receiving the feedback signal, participants report whether they expect to outperform their matched peer. This allows us to cross feedback content (above, below, none) with prior beliefs (optimistic, pessimistic), yielding six mutually exclusive feedback types. The most informative contrast — and the paper's sharpest result — is between *positively reassured* participants (told above, predicted above; $n = 42$) and *negatively surprised* participants (told below, predicted above; $n = 37$). Both groups shared identical optimistic priors, but their feedback either confirmed or contradicted those beliefs. The confirmed group scores 1.16 points higher on effort attribution ($p < 0.05$) and redistributes \$0.63 less ($p < 0.05$), consistent with motivated reasoning reinforcing positive self-views when feedback aligns with them. The main limitation is that near-universal overconfidence (85% predicted above) concentrates the positive-surprise cell at just $n = 5$, preventing estimation of the symmetric case and limiting power for the within-arm comparisons more broadly. A design with more balanced prior expectations would be needed to fully map the interaction between prediction direction and feedback valence.

The rest of the paper is structured as follows. Section 2 reviews the theoretical and experimental background. Section 3 describes the experimental design. Section 4 presents the results. Section 5 concludes.

2 Theoretical and Experimental Background

This paper builds on four overlapping strands of literature: redistribution preferences under belief uncertainty, the accountability principle and luck–effort attribution, competitive experience in impartial-spectator designs, and self-serving belief updating. The experimental design follows Almås et al. (2020) and is motivated by the research questions of Cassar and Klein (2019) and Deffains et al. (2016).

Redistribution preferences and beliefs about inequality. A large experimental literature has studied redistribution preferences, identifying self-interest, social preferences, and social identity as key drivers (Alesina and Giuliano, 2011). A central theoretical insight, developed by Piketty (1995) and Bénabou and Tirole (2006), is that beliefs about whether inequality reflects effort or luck are among the strongest determinants of redistribution demand. Alesina and Angeletos (2005) provide cross-country evidence: Americans attribute poverty more to individual failure than Europeans do, a difference that partly explains the gap in redistribution policies between the two regions.

A related strand investigates how uncertainty about one's relative income position shapes redistribution preferences. Karadja et al. (2017) show that informing survey respondents of their true rank in the income distribution changes their preferences: individuals who discover they are wealthier than they initially believed become less supportive of redistribution. This survey evidence motivates our laboratory approach, in which we exogenously vary the comparative performance signal while holding actual performance constant, enabling cleaner causal inference.

The accountability principle and self-serving attribution. Konow (2000) establishes the accountability principle: individuals support redistribution to compensate for outcomes driven by luck and to reward effort. Building on this framework, Cappelen et al. (2007) and Cappelen et al. (2013) classify spectators into fairness types—libertarian, meritocratic, and egalitarian—and find that most individuals prefer some redistribution away from luck inequality, even in the absence of material stakes. Our redistribution task follows this spectator design, adapting the merit scenario from Almås et al. (2020).

The accountability principle is systematically distorted when individuals have a stake in the outcome. Babcock and Loewenstein (1995) document that parties in bilateral bargaining inflate their assessments of a fair settlement in a self-serving direction. In the redistribution domain, Deffains et al. (2016) demonstrate a *political* self-serving bias: individuals who succeed at a task are more likely to attribute their outcome to effort and to oppose redistribution, while those who fail attribute their outcome to luck and become more redistributive. Their analysis shows that this cognitive distortion can generate systematic class-based divergence in fairness views, since wealthier individuals have disproportionately experienced success. Because our impartial-spectator design eliminates monetary stakes for spectators, any analogous bias in our data cannot be purely interest-driven; it must operate through the cognitive channel theorised by Kunda (1990): motivated reasoning that aligns perceived facts about deservingness with one's own experience of performance.

Competitive experience and impartial-spectator designs. A growing experimental literature examines how winning or losing in competitive settings shapes distributive preferences. Cassar and Klein (2019), whose research question directly inspires this paper, show that framing performance as a failure relative to a counterfactual produces more luck-oriented beliefs and greater redistribution support, even when spectators bear no material stakes in the outcome. Our design extends their insight from counterfactual framing to *randomly assigned comparative feedback*, providing cleaner identification of the causal effect of relative standing information.

The broader impartial-spectator paradigm—from which we adapt our design following Almås et al. (2020)—has been used extensively to elicit fairness preferences free of material self-interest and to compare them across populations and contexts (Cappelen et al., 2007, 2013). The absence of monetary stakes for spectators rules out opportunistic fairness reasoning and isolates the belief-updating channel (Babcock and Loewenstein, 1995).

Our contribution: isolating the comparative signal. Prior studies, including Deffains et al. (2016) and Cassar and Klein (2019), document that performance feedback shifts attribution and redistribution. However, in those designs participants automatically observe their performance or receive implicit comparative feedback, making it difficult to separate the informational content of the comparative signal from the mere experience of task completion.

We address this gap by introducing a **no-feedback arm**: a fourth group completes the identical task and makes the same redistribution decision, but receives no information about their score or their relative standing against a peer. Unlike prior studies, where participants automatically see their performance, this arm allows us to compare those who merely completed the task with those who additionally received a directional comparative signal, attributing any difference specifically to the *information content* of being told one outperformed or underperformed.

In addition, before feedback is revealed, participants state their expected relative performance. This pre-feedback elicitation allows us to classify participants by whether the feedback *confirmed* or *contradicted* their prior belief, and to test whether surprising feedback triggers stronger belief updating. The sharpest contrast in the data—between those whose optimistic prior was confirmed (*positively reassured*) and those whose optimistic prior was contradicted (*negatively surprised*)—provides direct evidence of a confirmation channel consistent with motivated reasoning models in which positive self-views are reinforced more readily than they are revised downward (Bénabou and Tirole, 2006; Kunda, 1990).

3 Experimental Design

The experiment was conducted online using JATOS and hosted the free, ESCoP-sponsored server Mindprobe. The whole design builds on the spectator design of Almås et al. (2020). It involved two distinct roles: *workers* and *spectators*.

Workers ($n = 2$) were recruited through Prolific and completed the code-recognition task under real monetary incentives. They were paid a base fee of \$4 plus a performance-based bonus of up to \$8, for a maximum possible payout of \$12 each. They were informed in advance that their earnings would initially depend on their relative performance against each other—the higher-scoring worker would receive \$8 while the other would receive \$0—and that a third party (a spectator) would subsequently decide whether and how much to redistribute between them.

Spectators ($n = 134$) were recruited through personal networks (friends, family, and colleagues from two workplaces), completed the full experiment online in approximately 10 minutes, and received no monetary compensation. The experiment consisted of two parts: a code-recognition task and a redistribution decision task in which they acted as impartial spectators. Spectators were informed before making their redistribution decisions that the workers were real individuals who had completed the same task and would receive actual monetary payouts determined by one randomly selected spectator's allocation. The impartial-spectator design thus removes material self-interest on the spectator side while keeping real stakes for the workers, ensuring that decisions have genuine consequences and giving the design greater external validity.

3.1 Part 1: Code-Recognition Task

Participants (both workers and spectators) completed a code-recognition task consisting of one practice round of 20 seconds followed by five scored rounds of 45 seconds each. At the start of each round, a three-digit target number (drawn from 100–999) was displayed at the top of the screen. Below it, a grid of three-digit numbers appeared; approximately 15 percent of the codes in the grid matched the target (around 48 instances per round), and the rest were distractors. Participants had to identify and check every instance of the target as quickly as possible. The grid was generated randomly at the start of each round, so the target and distractor codes changed every 45 seconds. To ensure a consistent interface experience, the grid dimensions were adapted to screen size: desktop users saw a 20×16 grid (320 codes), while mobile users saw a 7×46 grid (322 codes), with the same number of target instances in both cases. Participants were explicitly told about the scoring system: they earned 1 point for each correct identification, lost 1 point for each false alarm, and received no penalty for a missed target. The maximum possible score across the five rounds was around 200 points; sample means range from 90 to 95 points across treatment groups (Table 1).[2]

Just after finishing, participants answered three pre-feedback self-report questions: (i) effort exerted (slider, 0–10); (ii) guessed score (numeric, range −200 to 200); and (iii) relative performance prediction ("I would perform better than or the same as the other participant" vs. "I would perform worse").

3.2 Treatments: Relative Performance Feedback

After the self-report, participants were randomly assigned to one of three feedback conditions.

1. **No feedback.** Participants were informed they could proceed to Part 2 without seeing their score or any comparison information.

2. **Above.** Participants were told: "You performed above your randomly matched peer."

3. **Below.** Participants were told: "You performed below your randomly matched peer."

The feedback signal was generated through a dynamic matching procedure. Before the main data collection, ten pilot participants completed the task, providing an initial distribution of scores and establishing a current best and worst performer. As each subsequent spectator completed the task, those randomly assigned to the *above* condition were matched with the current lowest scorer in the spectator pool; those assigned to the *below* condition were matched with the current highest scorer. Reference scores were updated continuously as the experiment progressed. Note that spectators were compared with other spectators in this pool and the matching was indeed random. This design guarantees genuine, non-deceptive feedback: every participant in the above condition truly outperformed their assigned reference peer, and every participant in the below condition truly underperformed theirs. Participants who could not receive their assigned treatment — above-condition participants who scored below the current worst, or below-condition participants who scored above the current best — were excluded from the analysis. At debriefing, participants were shown their actual score and informed of the matching procedure, so they could understand why they had received the message they did.

[2]This marked the end of the task for the two workers. Before exiting, they were reminded that one spectator, chosen at random, would decide how to split their earnings, and that this decision would be paid out as their bonus. Spectators continued to Part 2, as described bellow.

3.3 Part 2: Distribution Decisions

After the (non)feedback screen, all participants acted as impartial spectators and made a redistribution decision adapted from Almås et al. (2020).

Redistributive Decision in a Merit Scenario

Participants read that two real participants — the two workers who had previously completed the same task — were identified as Worker A and Worker B. Worker A had received a larger initial payment because they performed better; Worker B received nothing because they performed worse. Participants were asked to decide whether and how to redistribute the given endowment between the two workers. The decision was presented as a set of discrete options covering the full range from keeping full inequality to reversing it.

After the redistribution decision, participants completed a short demographics survey (age, gender, and education) followed by the key belief measure of this study. They were asked to respond to the following question using a slider:

> *"To what extent do you think the initial income (before any redistribution) of the participants in the previous task was determined by effort versus luck?"*

The slider ranged from 0 ("Completely determined by luck") to 10 ("Completely determined by effort"). This attribution to effort versus luck is the hypothesised mediator between relative performance feedback and redistribution preferences.

3.4 Key Outcome and Mediator Variables

- **Redistribution (merit):** USD allocated to Worker B in the merit scenario. Primary outcome variable.
- **Effort–luck belief:** Score on the 0–10 slider described above. Hypothesised mediator.

3.5 Sample

Spectators were recruited through personal networks: friends, family, and employees at two workplaces. The analysis sample consists of $N = 143$ spectators after excluding non-engagers (zero score and zero clicks). The sample is predominantly Colombian (76.5%), with the remainder drawn from other countries, primarily in Europe. As this is a convenience sample, findings should be interpreted with caution regarding external validity and may not generalise to other populations in terms of magnitudes; treatment assignment was, however, random within this sample, so internal validity is preserved.

Table 1 presents descriptive statistics by treatment group. Groups are balanced on all pre-determined covariates: age ($p = 0.97$), gender ($p = 0.51$), task score ($p = 0.64$), nationality ($p = 0.42$), and education ($p = 0.48$), confirming that randomisation was successful. The nationality variable shows the above group with a slightly higher share of Colombians (81%) compared to the no-feedback (69%) and below (77%) groups, though this difference is comfortably non-significant ($p = 0.42$) and attributable to chance given the small sample size.

The effort–luck belief score differs significantly across groups ($p = 0.041$), which is itself a key outcome; importantly, this measure is elicited *after* the feedback treatment, so it cannot confound treatment assignment.

Table 1: Descriptive Statistics by Treatment Group

	Told: Above	No Feedback	Told: Below	p-value
Task score	91.26 (17.63)	94.84 (20.77)	91.06 (26.89)	0.640
Age	34.87 (13.62)	35.39 (14.78)	34.70 (14.67)	0.971
Female (%)	59.57 (49.61)	61.22 (49.23)	70.21 (46.23)	0.513
Colombian (%)	80.85 (39.77)	69.39 (46.57)	76.60 (42.80)	0.417
Predicted above average (%)	89.36 (31.17)	87.76 (33.12)	78.72 (41.37)	0.290
Education (%)				0.484
Less than bachelor	23.40 (42.80)	18.37 (39.12)	10.64 (31.17)	
Bachelor's	48.94 (50.53)	48.98 (50.51)	61.70 (49.14)	
Master's or above	27.66 (45.22)	32.65 (47.38)	27.66 (45.22)	
N	47	49	47	143

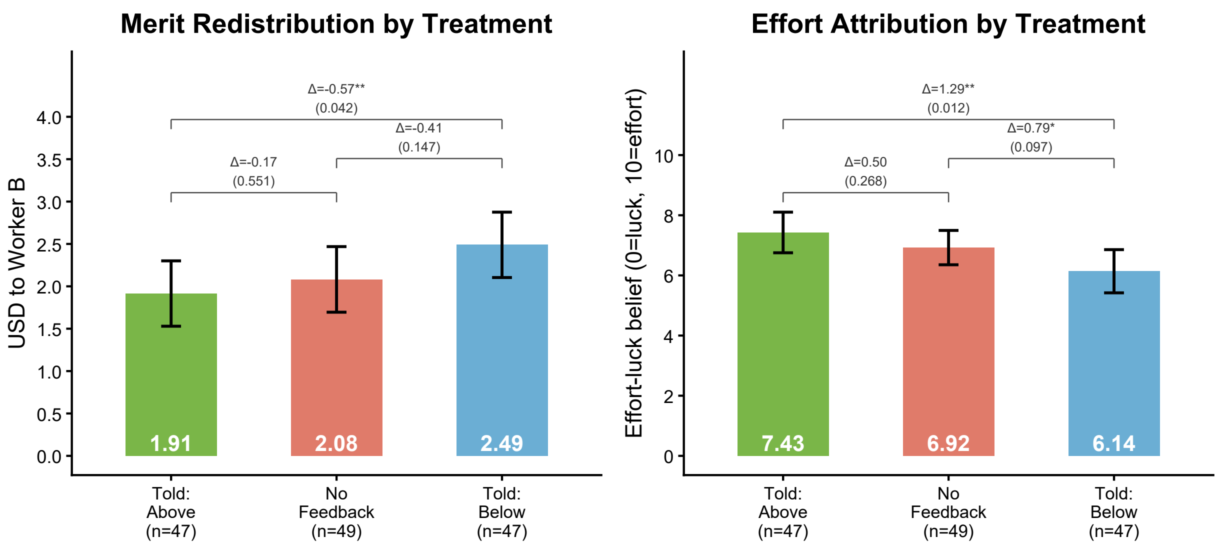
Mean (SD) for continuous variables; mean × 100 (SD × 100) for binary. p-values from one-way ANOVA (continuous) or chi-squared test (binary). Predicted above average: share expecting to outperform matched peer, elicited before feedback.

4 Results

Figure 1 gives an overview of both main outcomes across the three treatment arms. Among the 143 spectators in the sample, 47 were told they outperformed their matched peer (above), 47 were told they underperformed (below), and 49 received no feedback. Both panels display the same monotone gradient: the above group redistributes least and attributes most to effort, the below group is most redistributive and most luck-oriented, and the no-feedback group sits between the two, serving as an uninformed baseline anchor. Brackets show Welch t-test p-values for all three pairwise comparisons within each panel.

The sharpest contrast across both outcomes is the above–below gap, which is significant

Figure 1: Merit Redistribution and Effort Attribution by Treatment Group



Notes. Left panel: mean USD allocated to the lower-earning worker (merit scenario). Right panel: mean effort–luck belief (0 = 100% luck; 10 = 100% effort), elicited after the redistribution decision. Bar heights are group means; whiskers are 95% confidence intervals (±1.96 SE). Brackets show pairwise Welch t-tests (two-sided); p-values in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. $N = 143$.

in both redistribution and effort beliefs ($p < 0.05$). The no-feedback arm consistently falls between the two extremes, suggesting that the *direction* of the comparative signal — not mere task completion — is what shifts outcomes. With the present sample size, statistical power to detect all pairwise gaps is limited. For redistribution, neither treated group individually differs from the no-feedback anchor at conventional significance levels. For effort beliefs, the signal is sharper: the below group scores significantly lower than the no-feedback group ($p < 0.10$), reflecting a stronger belief-shifting effect of negative feedback, while the above–no-feedback gap does not reach conventional significance. Across all three pairs, the p-values for effort belief differences are uniformly smaller than the corresponding redistribution p-values, consistent with beliefs being more precisely identified by the feedback signal — a pattern that supports the mediation interpretation developed in Section 4.3.

4.1 Redistribution in the Merit Scenario

The left panel shows mean redistribution following the same gradient. The above group allocates the least (\$1.92, SD = \$1.35), followed by the no-feedback group (\$2.08, SD = \$1.39) and the below group (\$2.49, SD = \$1.38). The one-way ANOVA yields $p = 0.179$, so initially we cannot reject equality of means across all three groups.

Columns 1–2 of Table 2 report the main OLS regressions ($N = 143$). Without controls (column 1), the below group redistributes \$0.57 more than the above group (SE = \$0.28, $p < 0.05$), while the no-feedback group redistributes \$0.17 more (SE = \$0.28), a difference that is not statistically significant. With the full set of controls (column 2), the below-group estimate is \$0.65 (SE = \$0.30, $p < 0.05$).

The above–below gap of \$0.57–0.65 is stable across specifications and economically meaningful, representing roughly 30–34% of the above-group mean of \$1.92, and is significant at the 5 percent level in both columns. A linear hypothesis test cannot reject that the no-feedback and below groups redistribute the same amount ($p = 0.26$).

4.2 Effort–Luck Beliefs

The right panel shows a clear monotone pattern: the above group scores highest ($\overline{EL}_{\text{above}} = 7.43$, SD = 2.36), followed by the no-feedback group ($\overline{EL}_{\text{nofb}} = 6.93$, SD = 2.03) and the below group ($\overline{EL}_{\text{below}} = 6.14$, SD = 2.57). Notably, all three groups score above the scale midpoint of 5, indicating that effort attribution is the modal view regardless of treatment; what the feedback shifts is the *degree* to which participants see the outcome as effort-determined. A one-way ANOVA rejects equality of group means ($p = 0.041$).

Columns 3–4 of Table 2 report OLS regressions of the effort–luck belief on treatment dummies, with the above group as the baseline. Without controls (column 3), the below group scores 1.29 points lower than the above group (SE = 0.51, $p < 0.05$). The no-feedback group scores 0.50 points lower (SE = 0.46), roughly half the magnitude of the below-group gap and not statistically significant. Taking the no-feedback group as a natural anchor, the point estimates suggest that positive feedback elevates effort attribution by roughly 0.50 points above this anchor, while negative feedback depresses it by approximately 0.79 points below it — both signals moving in the expected direction. The upward shift (above vs. no-feedback) is not statistically significant; the downward shift (below vs. no-feedback) is marginally significant ($p < 0.10$, Welch t-test), suggesting that negative feedback shifts merit-oriented beliefs more forcefully than positive feedback. Once controls are added (column 4), both estimates remain stable: the below-group gap widens slightly to 1.36 points (SE = 0.51, $p < 0.01$), and the no-feedback coefficient is -0.46 (SE = 0.45), confirming that the pattern does not reflect pre-existing compositional differences across groups. A direct Welch t-test comparing the below and no-feedback groups is marginally significant ($p < 0.10$, visible in Figure 1); the OLS linear restriction on the regression coefficients yields $p = 0.12$, consistent with the pairwise result given the small sample.

These findings are consistent with the motivated-reasoning account of Kunda (1990): positive comparative signals are more readily internalised because they serve the motivated goal of maintaining a positive self-image, while negative signals invite counter-argument and discounting (Mezulis et al., 2004). Because the comparative signal is exogenously assigned and uncorrelated with actual task performance, these belief differences cannot be explained by genuine differences in effort across groups, providing experimental support for the belief-updating channel theorised by Deffains et al. (2016).

4.3 The Belief Channel: Mediation Analysis

The results in the previous two sections reveal a strong correlation between effort–luck beliefs and redistribution: each one-point increase on the 0–10 belief scale is associated with \$0.20 less redistributed (SE = \$0.05, $p < 0.01$), consistent with the accountability framework (Konow, 2000; Cappelen et al., 2007). Table 3 uses this relationship to assess whether beliefs are the channel linking feedback to redistribution. Causal identification of mediation requires not only that the treatment is randomly assigned (which it is) but also that no omitted variable simultaneously affects both the mediator and the outcome — a condition that cannot be guaranteed here, since only the comparative signal is randomised, not beliefs themselves.

Two features of the data nonetheless point toward beliefs as the primary channel. Theoretically, comparative signals shift beliefs about the relative contribution of effort and luck, and it is those updated beliefs — not the signal per se — that alter redistribution preferences (Deffains et al., 2016; Kunda, 1990). Empirically, the treatment effect is estimated more precisely for beliefs than for redistribution ($t = 2.54$ vs. $t = 2.04$): a common omitted variable driving both outcomes equally would not naturally produce this ordering.

Consistent with this interpretation, once effort–luck belief is included as a covariate, the below-group redistribution coefficient falls from \$0.57 (column 1 of Table 2) to \$0.32 (SE = \$0.28), a reduction of approximately 45 percent; the no-feedback coefficient falls from \$0.17 to \$0.07 (SE = \$0.27). These results are best read as suggestive evidence that attributional beliefs are an important mechanism linking performance feedback to redistribution — not as definitive causal proof of the channel.

Table 2: Treatment Effects on Effort Attribution and Redistribution

	USD Redistributed		Effort Beliefs	
	(1)	(2)	(3)	(4)
Constant	1.915***	2.692***	7.428***	7.086***
	(0.199)	(0.989)	(0.348)	(1.456)
No Feedback	0.167	0.158	−0.503	−0.464
	(0.282)	(0.292)	(0.456)	(0.446)
Told: Below	0.574**	0.648**	−1.289**	−1.361***
	(0.281)	(0.301)	(0.508)	(0.514)
Age		−0.003		0.017
		(0.012)		(0.018)
Female		−0.205		0.255
		(0.242)		(0.404)
Colombian		−0.305		0.324
		(0.299)		(0.520)
Task score		0.001		−0.011
		(0.008)		(0.012)
Bachelor's degree		−0.506		0.461
		(0.363)		(0.560)
Master's or above		−0.481		0.505
		(0.402)		(0.595)
N	143	143	143	143
R^2	0.031	0.071	0.051	0.107
Demographic controls	No	Yes	No	Yes

* p <0.1, ** p <0.05, *** p <0.01

Notes. OLS with HC3-robust SEs in parentheses. Baseline: Told Above. Cols. (1)–(2): USD redistributed to lower-earning worker (merit scenario). Cols. (3)–(4): effort–luck belief (0 = luck, 10 = effort). Controls: age, female indicator, Colombian indicator, task score, and education (Less than bachelor = baseline). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 3: Mediation Analysis

	(1) Merit redistrib. ($)	(2) Merit redistrib. ($)
Constant	3.407***	4.030***
	(0.432)	(0.956)
No Feedback	0.066	0.070
	(0.272)	(0.282)
Told: Below	0.315	0.391
	(0.275)	(0.293)
Effort–luck belief	−0.201***	−0.189***
	(0.051)	(0.053)
Age		0.000
		(0.011)
Female		−0.157
		(0.231)
Colombian		−0.244
		(0.292)
Task score		−0.001
		(0.007)
Bachelor's degree		−0.419
		(0.339)
Master's or above		−0.385
		(0.376)
N	143	143
R^2	0.143	0.164
Demographic controls	No	Yes

* p <0.1, ** p <0.05, *** p <0.01

Notes. OLS with HC3-robust SEs in parentheses. Dependent variable: USD redistributed (merit scenario). Includes effort–luck belief as control to assess mediation; compare treatment coefficients with Table 2 cols. (3)–(4). Baseline: Told Above. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

4.4 Pre-Feedback Expectations and the Surprise Channel

A novel element of this design is that participants state their expected relative performance *before* receiving the feedback signal. After the code-recognition task but before learning whether they performed above or below their peer, participants were asked to predict whether they would outperform the matched participant. This allows us to classify participants by whether the feedback was *surprising* — that is, whether it contradicted their prior expectation — and to ask whether unexpected feedback triggers stronger belief updating and redistribution responses.

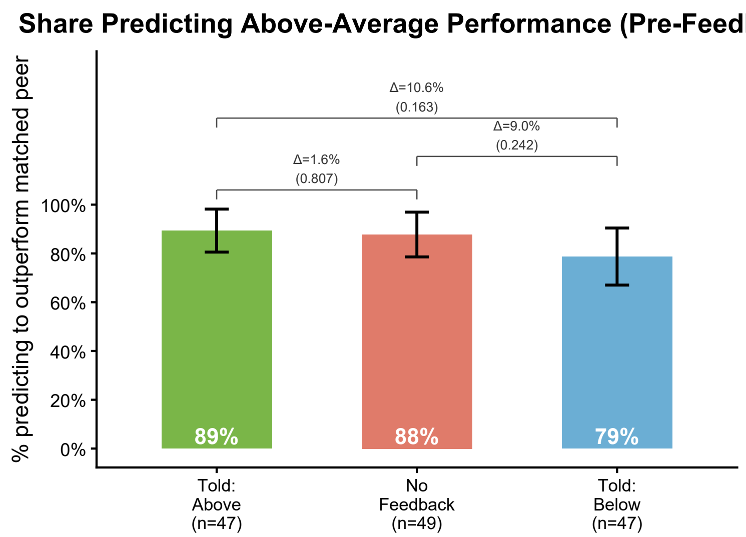


Figure 2: Pre-Feedback Performance Predictions by Treatment Group

Notes. Share of participants predicting they would outperform their matched peer, elicited before the feedback signal was revealed. Bar heights are group means; whiskers are 95% confidence intervals (±1.96 SE). Brackets show pairwise Welch t-tests (two-sided); p-values in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. $N = 143$.

Figure 2 shows the distribution of pre-feedback predictions by treatment group. The large majority of participants expected to outperform their peer: 89% in the above group, 88% in the no-feedback group, and 79% in the below group (85% overall). This pattern reflects the well-documented optimism bias or "better-than-average" effect (Mezulis et al., 2004): most individuals expect above-average performance even when only half can logically do so.

To isolate the surprise component, Table 4 compares outcomes within each treatment arm by whether the feedback contradicted the participant's prior expectation. Columns (1) and

(4) restrict to the above group and regress each outcome on a dummy for being positively surprised (told above despite predicting below; $n = 5$). Columns (2) and (5) restrict to the below group and regress on a dummy for being negatively surprised (told below despite predicting above; $n = 37$). The constant in columns (1)–(2) and (4)–(5) is the mean for the *not-surprised* sub-group: those whose feedback confirmed their prediction.

None of the surprise coefficients are statistically significant. Among the above group, positively surprised participants score slightly *lower* on effort attribution (-0.50, SE = 1.72) and redistribute slightly *more* (\$0.32, SE = \$0.77) than the not-surprised above participants — the direction one might expect if an unexpected win feels more "lucky" than a confirmed one, though the estimates are far too imprecise to draw conclusions ($n = 5$). Among the below group, negatively surprised participants score slightly *higher* on effort attribution (+0.86, SE = 0.93) and redistribute slightly more (\$0.11, SE = \$0.53) than the not-surprised below participants. A notable pattern is that the not-surprised below group (those who expected to lose and did; $n = 10$) shows the lowest effort attribution of all sub-groups (5.46 on the 0–10 scale), suggesting that individuals who are consistently pessimistic about their own performance also hold stronger luck-oriented beliefs.

Columns (3) and (6) pool all 143 observations and regress effort attribution and redistribution on a six-category factor variable for feedback type, with the *negatively surprised* group (told below, predicted above; $n = 37$) as the baseline. The remaining five categories are: *negatively reassured* (told below, predicted below; $n = 10$), *positively surprised* (told above, predicted below; $n = 5$), *positively reassured* (told above, predicted above; $n = 42$), *no feedback – optimistic* (no feedback, predicted above; $n = 43$), and *no feedback – pessimistic* (no feedback, predicted below; $n = 6$).

Two groups differ significantly from the negatively surprised baseline. First, *positively reassured* participants score 1.16 points higher on effort attribution (SE = 0.55, $p < 0.05$) and redistribute \$0.63 less (SE = \$0.31, $p < 0.05$): those whose optimism was confirmed by the feedback are the most merit-oriented of all groups. Second, *no-feedback pessimists* score 1.35 points higher on effort attribution (SE = 0.70, $p < 0.10$) than the negatively surprised baseline, though their redistribution does not differ significantly ($-$\$0.68, SE = \$0.57). No other group differs significantly from the negatively surprised baseline in either outcome. Notably, the *no-feedback optimists* ($n = 43$), who represent the majority of the no-feedback arm, are statistically indistinguishable from the negatively surprised baseline, which explains why the no-feedback treatment coefficient in Table 2 closely mirrors the below coefficient rather than the above.

The divergence between no-feedback pessimists and optimists is intriguing: pessimistic participants who receive no comparative signal still hold effort-oriented beliefs comparable to the positively reassured group, while optimistic no-feedback participants do not. This suggests that pre-existing beliefs about one's relative standing interact with feedback reception in ways that a single no-feedback dummy cannot capture, and warrants further investigation with larger samples.

Table 4 underscores the main limitation of the within-arm analysis: the R^2 values in columns (1)–(4) are all below 0.01, and the positive-surprise cell is too small ($n = 5$) to yield any useful estimate. The full-sample results in columns (3)–(6) further suggest that the treatment effect documented in Sections 4.2–4.1 is primarily driven by the contrast between those whose optimistic prediction was confirmed (*positively reassured*) and those whose optimistic prediction was contradicted (*negatively surprised*), rather than by the surprise element per se. Identifying the causal role of surprise and the no-feedback heterogeneity would require a sample with more balanced prior expectations and a larger pessimistic no-feedback cell.

5 Conclusion

This paper provides experimental evidence that relative performance feedback shapes beliefs about the role of effort in generating inequality and, through that channel, redistribution preferences. Participants told they outperformed their peer attribute significantly more of the task outcome to effort than those told they underperformed (gap of 1.29 points on a 0–10 scale, $p < 0.05$), and redistribute less in the merit scenario (gap of \$0.57–0.65, $p < 0.05$). A mediation analysis shows that approximately 45 percent of this redistribution gap operates through the shift in effort–luck beliefs, consistent with self-serving attribution as the operative channel; the uniformly smaller p-values for belief differences relative to redistribution support the interpretation that beliefs are upstream. The no-feedback arm anchors in the middle of both outcomes, consistent with the *direction* of the comparative signal — not mere task completion — being what drives the update: for redistribution, neither treated group differs significantly from the no-feedback anchor, while for effort beliefs the below group scores significantly lower than the no-feedback group ($p < 0.10$), reflecting a stronger belief-shifting effect of negative than positive feedback.

Beyond the main treatment effect, the pre-feedback performance prediction reveals a confirmation channel. Among participants who expected to outperform their peer, those whose expectation was confirmed (*positively reassured*) score 1.16 points higher on effort attribution and redistribute \$0.63 less than those whose expectation was contradicted (*negatively surprised*), both significant at the 5 percent level. This finding is consistent with motivated reasoning: comparative feedback amplifies the self-serving attribution effect most strongly when it aligns with the recipient's prior self-view, reinforcing positive self-views more readily

Table 4: Surprise Effects: Within-Arm and Full-Sample Comparisons

	USD Redistributed			Effort Beliefs		
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	1.881***	2.400***	2.514***	7.481***	5.460***	6.322***
	(0.210)	(0.477)	(0.224)	(0.354)	(0.833)	(0.420)
Pos. surprised (within above arm)	0.319			−0.501		
	(0.771)			(1.721)		
Neg. surprised (within below arm)		0.114			0.862	
		(0.527)			(0.932)	
Negatively reassured			−0.114			−0.862
			(0.527)			(0.932)
Positively surprised			−0.314			0.658
			(0.775)			(1.736)
Positively reassured			−0.633**			1.159**
			(0.308)			(0.549)
No feedback – optimistic			−0.397			0.499
			(0.313)			(0.532)
No feedback – pessimistic			−0.680			1.345*
			(0.569)			(0.696)
N	47	47	143	47	47	143
R^2	0.005	0.001	0.034	0.004	0.020	0.064

* p <0.1, ** p <0.05, *** p <0.01

Notes. OLS with HC3-robust SEs in parentheses. Cols. (1) and (4): above group only ($N = 47$); positively surprised = told above but predicted below ($n_{\text{surp}} = 5$). Cols. (2) and (5): below group only ($N = 47$); negatively surprised = told below but predicted above ($n_{\text{surp}} = 37$). Constant in cols. (1)–(2) and (4)–(5) = mean of the not-surprised sub-group within each arm. Cols. (3) and (6): full sample ($N = 143$); baseline = negatively surprised. Negatively reassured = told below, predicted below ($n = 10$); positively surprised = told above, predicted below ($n = 5$); positively reassured = told above, predicted above ($n = 42$); no feedback – optimistic = no feedback, predicted above ($n = 43$); no feedback – pessimistic = no feedback, predicted below ($n = 6$). Constant in cols. (3) and (6) = mean of the negatively surprised group. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

than it revises them downward.

Taken together, these findings support a self-serving attribution account of redistribution preferences: comparative performance signals shift merit-oriented beliefs, which in turn shift redistribution; and this shift is strongest when feedback *confirms* rather than challenges an existing self-view. The broader implication is that relative performance information is not a neutral signal — it interacts with prior beliefs in ways that can entrench meritocratic worldviews and reduce redistribution support, particularly among those who consistently perceive themselves as high performers. In a world where performance rankings are increasingly visible — on professional platforms, in academic settings, and on social networks — the self-serving dynamics documented here could contribute to a systematic polarisation of redistribution preferences between those who "win" and those who "lose" such comparisons.

Several limitations constrain the interpretation of these findings. First, although treatment assignment is random, the mediation analysis does not have causal force: effort–luck beliefs are not experimentally manipulated, so the estimated relationship between beliefs and redistribution may reflect unobserved characteristics correlated with both. Second, the sample is a convenience sample recruited largely through personal networks and is predominantly Colombian ($\approx 77\%$), limiting external validity; the magnitudes reported here may not generalise to other populations. Treatment assignment was, however, random within this sample and groups are balanced on all pre-determined covariates (Table 1), so internal validity is preserved. Third, near-universal overconfidence — 85 percent predicted to outperform — left very few pessimists in each arm and severely imbalanced the six feedback-type cells. The positive-surprise cell (told above, predicted below) contains only $n = 5$, making that contrast essentially uninformative. Even the main within-arm comparison — positively reassured ($n = 42$) versus negatively surprised ($n = 37$) — is constrained by the small overall sample: the modest size difference between these groups is the typical consequence of recruiting too few participants, not a design flaw.

Future work should address these limitations on several fronts. Most directly, replication with a larger sample would increase power for the individual contrasts with the no-feedback anchor, sharpen the within-arm surprise decomposition, and reduce cell imbalance across all six feedback types. A sample with more balanced prior expectations — achieved, for instance, by targeting populations where overconfidence is less prevalent or by framing the performance prediction differently — would allow clean identification of all six feedback $\times$ prediction combinations, including confirmed pessimists and positively surprised participants. To test the attribution channel causally, future designs could directly manipulate effort-versus-luck beliefs independently of performance feedback (for example, through vignette-based belief assignments), thereby isolating the mediating role of beliefs from other pathways. Finally, testing whether the effect replicates in more demographically diverse and representative samples would establish external validity and shed light on whether the self-serving dynamics documented here are universal or context-specific.

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Appendix

Table A2: Treatment Effects on Effort Attribution and Redistribution (Baseline: No Feedback)

	USD Redistributed		Effort Beliefs	
	(1)	(2)	(3)	(4)
Constant	2.082***	2.850***	6.924***	6.622***
	(0.199)	(0.974)	(0.294)	(1.401)
Told: Above	−0.167	−0.158	0.503	0.464
	(0.282)	(0.292)	(0.456)	(0.446)
Told: Below	0.408	0.490*	−0.786*	−0.897*
	(0.282)	(0.292)	(0.473)	(0.483)
Age		−0.003		0.017
		(0.012)		(0.018)
Female		−0.205		0.255
		(0.242)		(0.404)
Colombian		−0.305		0.324
		(0.299)		(0.520)
Task score		0.001		−0.011
		(0.008)		(0.012)
Bachelor's degree		−0.506		0.461
		(0.363)		(0.560)
Master's or above		−0.481		0.505
		(0.402)		(0.595)
N	143	143	143	143
R^2	0.031	0.071	0.051	0.107
Demographic controls	No	Yes	No	Yes

* p <0.1, ** p <0.05, *** p <0.01

Notes. OLS with HC3-robust SEs in parentheses. Baseline: No Feedback. Cols. (1)–(2): USD redistributed to lower-earning worker (merit scenario). Cols. (3)–(4): effort–luck belief (0 = luck, 10 = effort). Controls: age, female indicator, Colombian indicator, task score, and education (Less than bachelor = baseline). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A3: Mediation Analysis (Baseline: No Feedback)

	(1) Merit redistrib. ($)	(2) Merit redistrib. ($)
Constant	3.472***	4.100***
	(0.424)	(0.938)
Told: Above	−0.066	−0.070
	(0.272)	(0.282)
Told: Below	0.250	0.320
	(0.272)	(0.280)
Effort–luck belief	−0.201***	−0.189***
	(0.051)	(0.053)
Age		0.000
		(0.011)
Female		−0.157
		(0.231)
Colombian		−0.244
		(0.292)
Task score		−0.001
		(0.007)
Bachelor's degree		−0.419
		(0.339)
Master's or above		−0.385
		(0.376)
N	143	143
R^2	0.143	0.164
Demographic controls	No	Yes

* p <0.1, ** p <0.05, *** p <0.01

Notes. OLS with HC3-robust SEs in parentheses. Dependent variable: USD redistributed (merit scenario). Includes effort–luck belief as control to assess mediation; compare treatment coefficients with Table A2 cols. (1)–(2). Baseline: No Feedback. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A4: Surprise Effects: Within-Arm and Full-Sample Comparisons (Baseline: No Feedback – Optimistic)

	USD Redistributed			Effort Beliefs		
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	1.881***	2.400***	2.116***	7.481***	5.460***	6.821***
	(0.210)	(0.477)	(0.219)	(0.354)	(0.833)	(0.327)
Pos. surprised (within above arm)	0.319			−0.501		
	(0.771)			(1.721)		
Neg. surprised (within below arm)		0.114			0.862	
		(0.527)			(0.932)	
No feedback – pessimistic			−0.283			0.846
			(0.567)			(0.644)
Negatively surprised			0.397			−0.499
			(0.313)			(0.532)
Negatively reassured			0.284			−1.361
			(0.524)			(0.894)
Positively surprised			0.084			0.159
			(0.773)			(1.716)
Positively reassured			−0.235			0.660
			(0.304)			(0.481)
N	47	47	143	47	47	143
R^2	0.005	0.001	0.034	0.004	0.020	0.064

* p <0.1, ** p <0.05, *** p <0.01

Notes. OLS with HC3-robust SEs in parentheses. Cols. (1) and (4): above group only ($N = 47$); positively surprised = told above but predicted below ($n_{\text{surp}} = 5$). Cols. (2) and (5): below group only ($N = 48$); negatively surprised = told below but predicted above ($n_{\text{surp}} = 37$). Cols. (3) and (6): full sample ($N = 145$); baseline = no feedback – optimistic (no feedback, predicted above; $n = 43$). No feedback – pessimistic = no feedback, predicted below ($n = 7$); negatively surprised = told below, predicted above ($n = 37$); negatively reassured = told below, predicted below ($n = 11$); positively surprised = told above, predicted below ($n = 5$); positively reassured = told above, predicted above ($n = 42$). Constant in cols. (3) and (6) = mean of the no-feedback optimistic group. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A5: Treatment Effects on Effort Attribution and Redistribution (No Intercept)

	USD Redistributed		Effort Beliefs	
	(1)	(2)	(3)	(4)
Told: Above	1.915***	2.692***	7.428***	7.086***
	(0.199)	(0.989)	(0.348)	(1.456)
No Feedback	2.082***	2.850***	6.924***	6.622***
	(0.199)	(0.974)	(0.294)	(1.401)
Told: Below	2.489***	3.339***	6.138***	5.725***
	(0.199)	(0.956)	(0.370)	(1.382)
Age		−0.003		0.017
		(0.012)		(0.018)
Female		−0.205		0.255
		(0.242)		(0.404)
Colombian		−0.305		0.324
		(0.299)		(0.520)
Task score		0.001		−0.011
		(0.008)		(0.012)
Bachelor's degree		−0.506		0.461
		(0.363)		(0.560)
Master's or above		−0.481		0.505
		(0.402)		(0.595)
N	143	143	143	143
R^2	0.723	0.734	0.900	0.906
Demographic controls	No	Yes	No	Yes

* p <0.1, ** p <0.05, *** p <0.01

Notes. OLS without intercept (`-1`) with HC3-robust SEs in parentheses. Each treatment coefficient equals the conditional group mean. Cols. (1)–(2): USD redistributed to lower-earning worker (merit scenario). Cols. (3)–(4): effort–luck belief (0 = luck, 10 = effort). Controls: age, female indicator, Colombian indicator, task score, and education (Less than bachelor = baseline). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.